

BUSAN GIMHAE INTL, Korea, Republic of

WMO: 471530

Lat: **35.180N**Lon: **128.938E**Elev: **6**StdP: **14.69**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.0	23.2	-2.4	4.8	30.1	1.4	5.9	30.8	21.7	36.2	19.3	37.9	7.0	330	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.5	91.5	78.6	89.4	77.9	86.4	76.7	80.3	87.5	79.4	86.4	78.3	84.9	10.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.7	148.8	85.2	77.1	141.3	83.8	75.6	134.1	82.5	44.1	88.2	42.9	86.5	41.6	85.0	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.0	17.4	15.7	DB	15.6	94.7	3.0	2.9	13.4	96.7	11.6	98.4	9.9	100.0	7.7	102.1
			WB	12.7	82.1	2.7	2.0	10.7	83.5	9.1	84.7	7.6	85.8	5.6	87.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.8	36.3	40.1	47.4	56.7	65.4	71.9	78.6	80.7	73.0	63.1	51.1	39.9	(6)
	DBStd	15.90	5.59	6.01	5.62	5.10	4.14	3.18	4.33	3.92	4.36	5.21	6.53	6.12	(7)
	HDD50	1221	427	282	118	8	0	0	0	0	0	1	67	318	(8)
	HDD65	3728	891	698	547	251	46	1	0	0	1	96	418	779	(9)
	CDD50	4421	1	3	36	210	477	658	885	952	689	407	99	4	(10)
	CDD65	1454	0	0	0	3	58	208	420	487	240	38	0	0	(11)
	CDH74	11171	0	0	0	20	301	905	3530	4863	1399	152	1	0	(12)
	CDH80	3382	0	0	0	1	32	122	1119	1807	296	5	0	0	(13)
(14) Wind	WSAvg	7.3	7.2	7.4	7.8	8.1	7.7	7.5	7.6	7.4	7.3	6.6	6.2	6.7	(14)
(15) Precipitation	PrecAvg	57.0	1.1	1.8	3.0	4.9	5.6	7.0	11.5	10.6	6.1	2.4	1.9	1.1	(15)
	PrecMax	93.7	2.5	4.4	5.9	9.2	12.2	13.5	26.8	28.5	18.7	7.4	8.3	4.3	(16)
	PrecMin	32.4	0.0	0.0	0.3	1.1	1.8	1.8	3.2	2.2	0.9	0.0	0.0	0.0	(17)
	PrecStd	13.7	0.7	1.1	1.5	2.2	2.4	3.8	6.1	6.9	4.8	1.9	1.7	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.2	61.1	68.2	76.9	82.8	85.9	94.6	95.0	88.1	80.5	72.0	61.2	(19)
		MCWB	49.9	52.2	52.9	60.8	64.8	69.7	79.8	79.0	76.0	66.5	60.4	52.3	(20)
	2%	DB	52.2	57.6	64.3	73.0	80.3	82.7	91.2	91.8	85.8	77.4	68.3	57.3	(21)
		MCWB	43.0	48.9	52.8	58.4	63.7	69.1	78.8	78.6	73.9	65.2	58.2	48.3	(22)
	5%	DB	49.7	55.1	61.0	69.8	77.2	80.9	88.1	89.9	83.8	75.4	65.9	54.0	(23)
		MCWB	41.1	46.4	51.3	57.5	63.0	68.7	78.1	78.1	72.8	64.5	56.4	45.7	(24)
	10%	DB	46.5	51.7	58.8	67.6	75.0	79.0	86.0	88.0	80.9	73.3	62.7	51.5	(25)
		MCWB	39.4	43.0	50.2	56.8	62.4	68.5	77.4	77.7	71.0	62.7	54.6	43.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.4	57.3	59.4	66.1	70.6	75.7	81.9	82.4	79.0	72.5	64.5	55.2	(27)
		MCDB	55.8	60.6	63.0	70.0	75.8	80.5	89.3	88.8	85.5	76.2	68.3	58.9	(28)
	2%	WB	45.4	51.7	56.7	63.0	67.9	74.0	80.1	80.7	76.9	69.1	61.0	50.5	(29)
		MCDB	49.7	55.7	61.0	68.3	73.5	78.1	87.2	88.0	82.4	74.4	66.1	55.6	(30)
	5%	WB	42.6	46.8	53.7	60.4	66.3	72.5	79.2	79.8	75.2	66.5	58.4	47.1	(31)
		MCDB	47.2	52.7	59.6	66.2	72.8	76.9	86.1	87.0	80.2	73.1	64.1	52.6	(32)
	10%	WB	40.0	44.0	50.5	58.4	64.7	71.0	78.1	78.9	72.9	63.8	55.5	44.2	(33)
		MCDB	45.3	49.7	57.2	64.7	71.7	76.1	84.5	85.8	78.7	70.8	61.2	50.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.9	18.6	18.6	18.5	17.1	13.6	11.1	12.5	14.6	18.3	19.1	18.5	(35)
		MCDBR	21.2	22.2	21.8	22.5	21.6	17.4	14.3	15.2	16.6	19.7	21.6	22.1	(36)
	5% WB	MCWBR	14.3	14.9	13.7	12.6	10.6	8.4	6.9	6.5	7.8	10.2	13.2	15.1	(37)
		MCWBR	17.4	19.6	18.4	17.1	16.7	11.4	12.8	13.3	12.9	15.9	18.3	19.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.410	0.461	0.536	0.567	0.580	0.620	0.592	0.548	0.496	0.461	0.429	0.403	(40)
		taud	2.082	1.981	1.820	1.800	1.822	1.802	1.931	2.045	2.135	2.113	2.110	2.114	(41)
	Ebn at Noon		240	241	233	234	233	224	229	237	243	239	232	233	(42)
		Edh at Noon	40	50	63	68	67	69	60	53	46	43	39	37	(43)
(44) All-Sky Solar Radiation	RadAvg		852	1053	1370	1584	1740	1538	1457	1479	1238	1145	874	778	(44)
	RadStd		68	115	126	124	184	139	221	180	131	109	104	47	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	N/A

Nomenclature: See separate page